# Ananth Jonnavittula

San Francisco, CA
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## Summary

Robotics researcher with 8+ years of experience across academic and industrial domains, specializing in perception-driven manipulation, visual imitation learning, and generative AI for robotics. Proven track record deploying production-grade vision systems and publishing impactful research in reinforcement and imitation learning. Seeking to develop on-foundational AI/ML technologies.

## Technical Skills

**Languages**: Python, URScript, KUKA KRL
**Frameworks/Tools:** PyTorch, TensorFlow, OpenCV, ROS, Isaac Sim, Blender, TorchScript, GitHub
**Robotics Platforms:** Universal Robots, ABB, FANUC, KUKA, FrankaEmika, Fetch
**Research Interests:** Visual Imitation Learning, Robot Perception, Self-Supervised Learning, Human-Robot Interaction, Foundation Models

## Education

**Virginia Tech** **Aug 2020 – Aug 2024**
*PhD in Mechanical Engineering* *Blacksburg, VA*

**Worcester Polytechnic Institute** **Aug 2015 – May 2017**
*MS in Robotics Engineering* *Worcester, MA*

**SASTRA University** **Jun 2011 – May 2015**
*B.Tech in Electronics and Instrumentation* *Tanjore, India*

## Work Experience

**Senior Research Engineer** **Sep 2024 – Present**
*Samsung Research America* *Mountain View, CA*

- Developed VR based teleoperation systems for humanoid robots
- Designed universal teleoperation system stack for AI data collection
- Developed Nav2 based navigation for humanoid robots

**AI Researcher** **Dec 2024 – Sep 2025**
*Foundation Future Industries Inc* *San Francisco, CA*

- Developed Diffusion Policy based Imitation Learning policies for dexterous manipulation
- Created segmentation and pose estimation pipelines using DINOv2 and SAM2 for uncommon automotive parts
- Integrated NeRF-based SLAM pipelines for scene understanding in real world factories
- Engineered Nvidia Isaac Sim based simulation environment for custom humanoid robot

**Graduate Research Assistant** **Aug 2020 – Aug 2024**
*Virginia Tech* *Blacksburg, VA*

- Developed an imitation learning method that can learn long horizon tasks from a single video demonstration
- Implemented Soft Actor-Critic based Reinforcement Learning approaches for manipulation of YCB objects
- Engineered an inverse reinforcement learning based algorithm to learn from imperfect user demonstrations
- Spearheaded the development of a variational autoencoder based algorithm for imitation learning
- Conducted stability analysis for imitation learning methods in the context of shared autonomy

**Research and Development Intern** **Jun 2022 – Aug 2022**
*ABB Inc* *San Jose, CA*

- Implemented and analyzed methods that use stereo vision or RGBD images for instance segmentation
- Architected an instance segmentation algorithm for small parcels singulation using detectron2
- Devised a BlenderProc based pipeline to generate synthetic datasets for training models
- Engineered a pipeline to export trained models from Python to C++ using Torchscript

**Robotics/Vision Engineer** **Jun 2017 – May 2020**
*Parker Hannifin Corp.* *Haverhill, MA*

- Engineered an automated cell for palletizing over 100 different SKUs to reduce labor costs
- Designed and implemented an automated laser marker that doubled throughput in multiple manufacturing cells
- Developed a robotic cell using UR5 robot and cameras for part recognition and end capped filter
- Engineered and deployed a urethane end capping cell using FANUC robots

**Engineering Intern** **Jan 2017 – Apr 2017**
*Parker Hannifin Corp.* *Haverhill, MA*

- Conducted feasibility analysis on automated end capping using collaborative robots
- Programmed controllers for automated part feeding using vibratory feeders
- Implemented image recognition using Keyence CV-X series to detect orientation of parts

**Graduate Research Assistant** **Jan 2016 – May 2016**
*Worcester Polytechnic Institute* *Worcester, MA*

- Conducted analysis for range of motion and kinematic requirements for a 2 DOF hydro-muscle actuated leg
- Designed a coupling mechanism that locks the leg while maintaining pose in case of serious failure
- Orchestrated the development closed-loop control system for leg actuation using four pairs of hydro-muscles
- Developed controllers for hydraulics and coupling mechanism using Arduino microcontrollers

## Publications

**VIEW: Visual Imitation Learning with Waypoints** **2025**
*Springer Autonomous Robots*
***Ananth Jonnavittula***, *Sagar Parekh, and Dylan P. Losey*

**SARI: Shared Autonomy across Repeated Interaction** **2024**
*ACM Transactions on Human-Robot Interaction*
***Ananth Jonnavittula***, *Shaunak A. Mehta, and Dylan P. Losey*

**Communicating Robot Conventions through Shared Autonomy** **2022**
*International Conference on Robotics and Automation (ICRA)*
***Ananth Jonnavittula*** *and Dylan P. Losey*

**Here's What I've Learned: Asking Questions that Reveal Reward Learning** **2022**
*ACM Transactions on Human-Robot Interaction*
*Soheil Habibian,* ***Ananth Jonnavittula***, *and Dylan P. Losey*

**Learning to Share Autonomy Across Repeated Interaction** **2021**
*IEEE International Conference on Intelligent Robots and Systems (IROS)* *Best Student Paper Finalist*
***Ananth Jonnavittula*** *and Dylan P. Losey*

**I know what you meant: Learning human objectives by (under)estimating their choice set** **2021**
*IEEE International Conference on Robotics and Automation (ICRA)*
***Ananth Jonnavittula*** *and Dylan P. Losey*

## Patents

**Biologically inspired joints and systems and methods of use thereof** **2020**
*US010632626B2*